

Deep Snake (Paper-Faithful) in Context

Cross-method comparison on FLAME-3 and FLAME-1

CS269 Project — Deformable Models

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1 What this compares

We place the new paper-faithful **deep_snake_paper** (separately-trained CenterNet detector $\rightarrow$ octagon init $\rightarrow$ circular-convolution snake) against every other method on both datasets. “Deep Snake (simple)” is the original detector-free variant (U-Net coarse mask seeds the contour). All scores are mean IoU/Dice on binary masks; rows are sorted by IoU.

Two datasets, two definitions of “fire”:

- **FLAME-3** (Sycon Marsh): GT = paired thermal frame thresholded at $\geq 150^\circ\text{C}$. A smoke-occluded hot region, largely *invisible* as orange flame in RGB.
- **FLAME-1**: GT = hand-labeled *visible-flame* PNG masks. The label is the flame you can see in RGB.

All methods take RGB only as input.

2 FLAME-3 (thermal-defined fire)

Table 1: FLAME-3. Classical methods scored on 592 non-empty frames; deep methods on the 94-frame test split. “oracle” = contour initialized from the GT mask.

Method	Init / family	IoU	Dice
GAC	oracle (from GT)	0.365	0.499
Kass Snakes	oracle (from GT)	0.198	0.305
GAC (U-Net seed)	learned	0.150	0.211
U-Net	learned (RGB)	0.147	0.207
DALS	learned (RGB)	0.122	0.182
Deep Snake (simple)	learned (RGB)	0.032	0.032
Deep Snake (paper)	CenterNet $\rightarrow$ snake	0.032	0.032
Color floor (R-G)	none	0.004	0.007

On FLAME-3 the paper pipeline is *indistinguishable from the floor* (0.032, the empty-vs-empty score): its CenterNet detector trained to **mAP 0.000** and detected nothing, so the snake never received a box. The whole table sits low — oracle-init GAC (0.365) leads only because it is handed the GT location; every prior-free method, deep or classical, struggles because thermal-defined fire is not recoverable from RGB (in-fire pixels measure $R-G \approx 2.5$ vs. 1.9 background; color floor = 0.004).

3 FLAME-1 (visible-flame)

Table 2: FLAME-1. Deep methods + GAC(U-Net) scored on the 301-frame test split; color / oracle-GAC / oracle-Kass on all 2,001 frames (sample size in parentheses). Deep Snake (paper) detector trained 10 epochs; snakes 30 epochs.

Method	Init / family	IoU	Dice
U-Net	learned (RGB)	0.772	0.870
DALS	learned (RGB)	0.747	0.854
Deep Snake (simple)	learned (RGB)	0.658	0.792
GAC (U-Net seed)	learned	0.646	0.784
GAC	oracle (from GT, n=2001)	0.592	0.738
Color floor (R-G, n=2001)	none	0.473	0.625
Deep Snake (paper)	CenterNet → snake	0.050	0.080
Kass Snakes	oracle (from GT, n=2001)	0.028	0.052

On FLAME-1 every learned method jumps dramatically — even the *color floor* (0.473) beats every FLAME-3 method, because visible flame really is red/orange. U-Net leads (0.772). The detector-free Deep Snake (simple) is competitive at 0.658.

`deep_snake_paper`, however, lands near the bottom at **0.050** — and this is the report’s central comparison point.

4 Why the paper pipeline is last (on both datasets)

The decoupled detector→snake design is gated by detection, and that gate dominates the score:

- **FLAME-3:** detector mAP 0.000 $\Rightarrow$ no boxes $\Rightarrow$ floor (0.032). Completely detection-limited.
- **FLAME-1:** the detector now *works* (mAP 0.120, mAP@50 **0.430** after the learning-rate fix), but at 10 epochs it fires on only **24% of test frames**. On the other 76% the snake gets no box and outputs empty, dragging the mean to 0.050.
- **The snake itself is fine.** Conditioned on the detector firing, `deep_snake_paper` reaches IoU **0.206** — 4× its unconditional score — and its oracle-box validation IoU was 0.180. The refinement head works; recall is the bottleneck.

By contrast, Deep Snake (simple) seeds from a *dense* U-Net coarse mask, which fires on 100% of frames, so it never starves — explaining the 0.658 vs. 0.050 gap between two methods that share the same snake refinement core.

5 Takeaways

1. **Dataset dominates.** Visible-flame FLAME-1 lifts every learned method far above its FLAME-3 score; thermal-defined FLAME-3 is near-unsolvable from RGB without an oracle.
2. **Paper-faithful $\neq$ better here.** Faithfully decoupling detection from refinement *added* a failure mode — detector recall — that the detector-free variant avoids. Deep Snake (paper) is last on both datasets despite a sound snake head.
3. **The gap is recall, and it is closable.** The detector was capped at 10 epochs (runtime) and was still improving; FLAME-1’s fire boxes are also fragmented (median ~ 2 dominant components/frame). More detector training and less-fragmented box targets are the obvious next levers — the snake’s given-a-box performance (0.206) sets the headroom.

Sources: FLAME-3 + FLAME-1 baselines from `results/boundary_summary.csv` and `RESULTS.md`; Deep Snake (paper/simple) from `results/rerun_flame1_20260601_193349/` (FLAME-1) and the FLAME-3 rerun (`deep_snake_paper` 0.032, detector mAP 0). *Caveats:* (i) within FLAME-1, classical/oracle rows use all 2,001 frames while deep rows use the 301 test frames; (ii) oracle methods are handed the GT location and are not detection-comparable; (iii) the paper detector ran 10 epochs, below convergence.